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OF APPLIED SCIENCES

Electronic Circuit Design

Lecture notes

Chapter 3: Voltage and Current Sources

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Contents

3 Voltage and Current Sources **3**
3.1 Ideal and Real Voltage Sources 3
3.2 Current Sources and Source Transformation 5
3.3 The Zener Diode Shunt Regulator 7
3.4 Series Regulators 10
3.5 Op-Amp Voltage Sources 13
3.6 Transistor Current Sources 15
3.7 Op-Amp Current Sources and Sensor Excitation 18

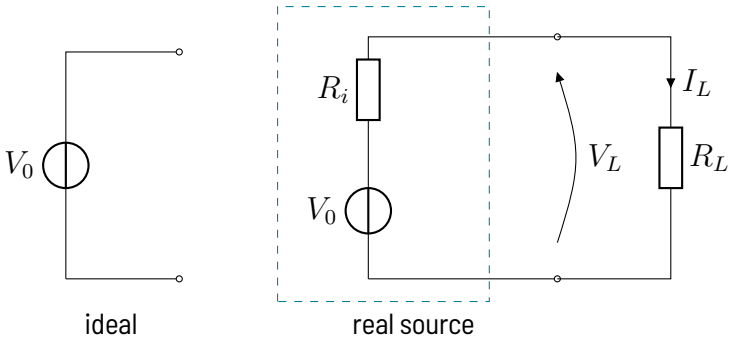
References **22**

3 Voltage and Current Sources

3.1 Ideal and Real Voltage Sources

Every circuit needs energy, and every measurement needs a quantity that is known exactly: a supply voltage, a reference voltage or an excitation current for a sensor. This chapter shows how real sources differ from the ideal ones of circuit theory, and how simple circuits with Zener diodes, transistors and op-amps produce stable voltages and currents. In the capstone task the bridge needs a stable supply, the reference ladder of the flash converter needs a stable 4 V, and the resistive sensor must not carry more than 1 mA. The chapter on voltage references continues with precise references (bandgap references); here we treat the basic circuits.

Ideal and real voltage source



- ideal: the voltage is V_0 for every current
- real: the terminal voltage drops with the load current

Terminal voltage of a real source

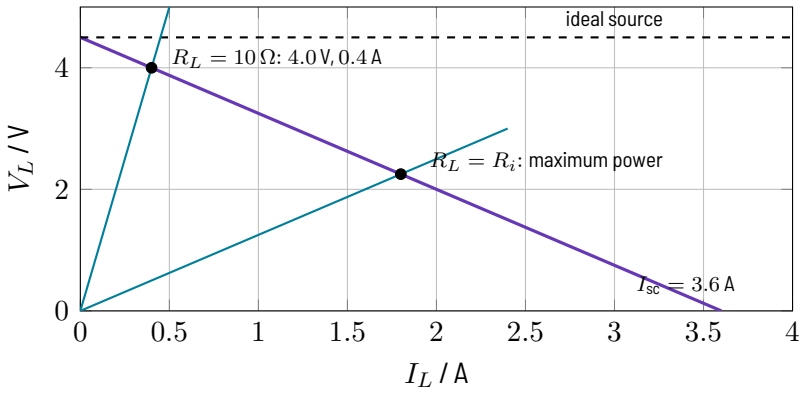
$$V_L = V_0 - R_i\, I_L = V_0\, \frac{R_L}{R_i + R_L}$$

Examples of internal resistances

- AA alkaline cell: $V_0 = 1.5\,\text{V}$, $R_i \approx 0.1\,\Omega \ldots 0.3\,\Omega$ (rises as the cell discharges)
- 9 V block battery: $R_i \approx 1\,\Omega \ldots 2\,\Omega$
- laboratory power supply in voltage mode: some mΩ, plus the leads
- voltage divider: $R_i = R_1 \parallel R_2$ (chapter Basics)
- logic output: $20\,\Omega \ldots 50\,\Omega$
- op-amp output with feedback: well below $1\,\Omega$ at low frequencies

The internal resistance is not a resistor one could find inside the source; it is a model for everything that makes the voltage drop under load: the chemistry of a battery, the control loop of a power supply, the output transistors of a logic gate. A real source is therefore described by two numbers, V_0 and R_i, which can be measured: the open-circuit voltage directly, the internal resistance from a second measurement with a known load, $R_i = (V_0 - V_L)/I_L$. A short circuit to measure I_{sc} directly is not a good idea for batteries and power supplies.

Load line and operating point

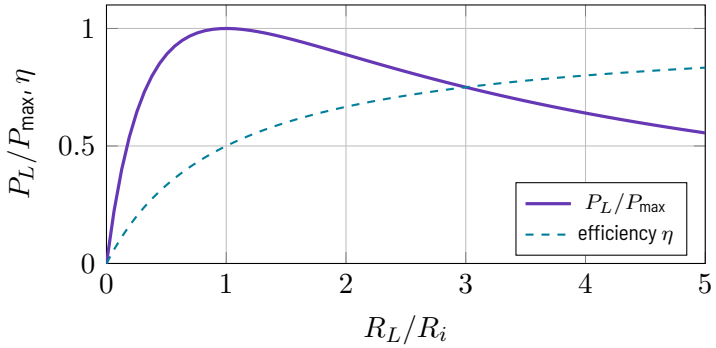


- example: $V_0 = 4.5\,\text{V}$, $R_i = 1.25\,\Omega$; load lines of $R_L = 10\,\Omega$ and $R_L = R_i$

- the operating point is the intersection of source line and load line

▶ Falstad: `examples/sources/real_source.txt`

Power matching



- load power $P_L = V_0^2\, R_L/(R_i + R_L)^2$ is maximum for $R_L = R_i$: $P_{\max} = V_0^2/(4R_i)$
- then half of the power is lost in the source: efficiency $\eta = 50\,\%$
- power supplies and signal sources for measurement: $R_L \gg R_i$ (voltage matching)

Power matching is important in radio frequency circuits and for antennas, where the cable impedance must be matched, and for weak energy sources such as thermoelectric generators. For supplies and in measurement circuits it is the wrong goal: we want the voltage to stay constant, so the load resistance must be large compared with the internal resistance. The same rule appeared for the loaded voltage divider.

Exercise 3.1: Measuring a real voltage source

A battery pack shows $4.5\,\text{V}$ without load and $4.0\,\text{V}$ with a load of $10\,\Omega$.

a) Calculate the load current and the internal resistance.

b) Calculate the short-circuit current.

c) Which load takes the maximum power from the battery? How large is this power and the efficiency?

d) Check (a) in Falstad and use the slider to find the maximum load power.

Solution 3.1

a) $I_L = 4.0\,\text{V}/10\,\Omega = 0.4\,\text{A}$, $R_i = (4.5\,\text{V} - 4.0\,\text{V})/0.4\,\text{A} = 1.25\,\Omega$.

b) $I_{\text{sc}} = V_0/R_i = 4.5\,\text{V}/1.25\,\Omega = 3.6\,\text{A}$.

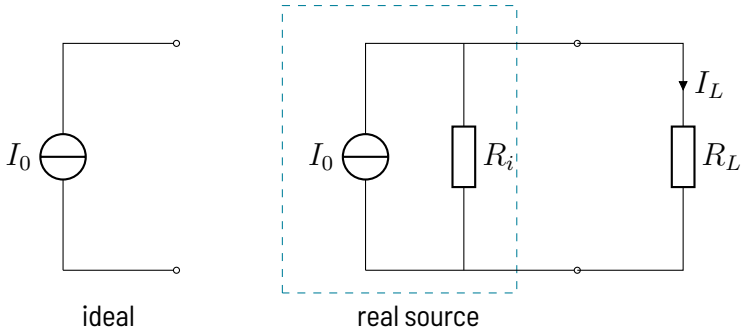
c) $R_L = R_i = 1.25\,\Omega$: $P_{\max} = (4.5\,\text{V})^2/(4 \cdot 1.25\,\Omega) = 4.05\,\text{W}$, $V_L = 2.25\,\text{V}$, $\eta = 50\,\%$; the battery heats up with the same $4.05\,\text{W}$.

d) ▶ Falstad: `examples/sources/real_source.txt`: `V_ideal` $= 4.50\,\text{V}$, `V_term` $= 4.00\,\text{V}$, $I_L = 0.400\,\text{A}$ (hover over R_L). With the slider at $R_L = 1.25\,\Omega$ the resistor shows $4.05\,\text{W}$.

3.2 Current Sources and Source Transformation

A current source delivers a given current whatever voltage the load needs. In the lab it is less familiar than a voltage source, but it is everywhere inside integrated circuits: it biases transistor stages, it charges capacitors linearly, and it excites resistive sensors, because the voltage across the sensor is then directly proportional to its resistance.

Ideal and real current source

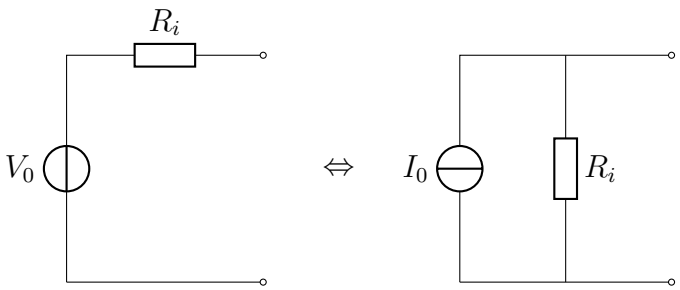


Load current of a real current source

$$I_L = I_0 - \frac{V_L}{R_i} = I_0 \frac{R_i}{R_i + R_L}$$

- a good current source has a **large** internal resistance ($R_i \gg R_L$)
- an ideal current source must never be left open: its voltage would rise without limit
- real current sources work only in a limited voltage range: the **compliance range**

Source transformation



- both have the same open-circuit voltage V_0 and the same short-circuit current $I_0 = V_0/R_i$
- therefore they behave identically for **every** load (Thevenin ↔ Norton)
- inside, they are different: the power lost in R_i is not the same
- example: $10\,\text{V}$ with $10\,\text{k}\Omega \Leftrightarrow 1\,\text{mA}$ with $10\,\text{k}\Omega$; with $R_L = 4.7\,\text{k}\Omega$ both give $3.20\,\text{V}$

▶ Falstad: `examples/sources/source_transform.txt` Whether a source is called a voltage source or a current source is a question of the load. A source with $R_i = 10\,\text{k}\Omega$ is a fairly good voltage source for a $1\,\text{M}\Omega$ load and a fairly good current source for a $100\,\Omega$ load. Source transformation is also a practical tool for calculations: series connections are easy with voltage sources, parallel connections with current sources.

Exercise 3.2: Source transformation

A source consists of $10\,\text{V}$ in series with $10\,\text{k}\Omega$. It feeds $R_L = 4.7\,\text{k}\Omega$.

a) Transform it into a current source with a parallel resistor.

b) Calculate the load voltage with both models.

c) Calculate the power dissipated in the internal resistance for both models. What do you conclude?

d) Check (b) in Falstad.

Solution 3.2

a) $I_0 = 10\,\text{V}/10\,\text{k}\Omega = 1\,\text{mA}$ parallel to $10\,\text{k}\Omega$.

b) Voltage source: $V_L = 10\,\text{V} \cdot 4.7/14.7 = 3.20\,\text{V}$. Current source: $V_L = 1\,\text{mA} \cdot (10\,\text{k}\Omega \parallel 4.7\,\text{k}\Omega) = 1\,\text{mA} \cdot 3.20\,\text{k}\Omega = 3.20\,\text{V}$.

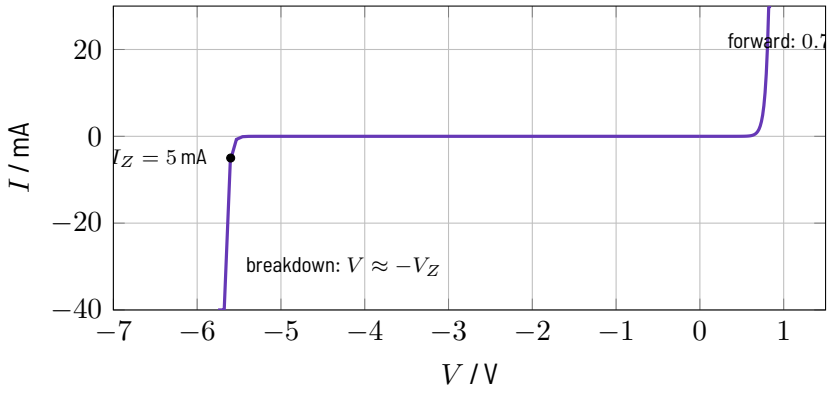
c) Voltage source: $I = 0.680\,\text{mA}$ flows through R_i, $P = 4.63\,\text{mW}$. Current source: R_i sees $3.20\,\text{V}$, $P = 1.02\,\text{mW}$. The models are equivalent only at the terminals, not inside.

d) ▶ Falstad: `examples/sources/source_transform.txt`: `VL_thevenin` = `VL_norton` = $3.20\,\text{V}$.

3.3 The Zener Diode Shunt Regulator

A Zener diode is a diode that is operated in reverse breakdown. Above the breakdown voltage the reverse current rises steeply, so the voltage stays almost constant over a wide current range. Together with a series resistor this gives the simplest voltage regulator. It is good enough for small currents and as a reference for the transistor regulator of the next section.

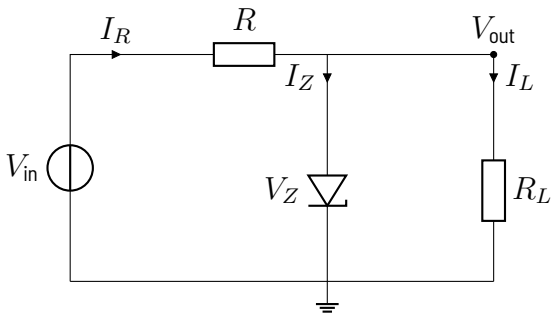
Zener diode characteristic



- breakdown voltages $2.4\,\text{V}$ … $75\,\text{V}$ in E24 steps (BZX55, BZX79: $0.5\,\text{W}$; 1N47xx: $1.3\,\text{W}$)
- datasheet: V_Z at a test current (often $5\,\text{mA}$), dynamic resistance $r_Z = \Delta V / \Delta I$

Below about $5\,\text{V}$ the breakdown is caused by the Zener effect (tunnelling), above by avalanche multiplication. Diodes around $5\,\text{V}$ … $6\,\text{V}$ have the smallest temperature coefficient and the smallest dynamic resistance, which is why $5.6\,\text{V}$ and $6.2\,\text{V}$ are popular reference values. Low voltage Zener diodes have a soft knee: a $2.7\,\text{V}$ type has a dynamic resistance of some $100\,\Omega$. The Falstad model is calibrated so that the current is exactly $5\,\text{mA}$ at the nominal V_Z, and it is much stiffer than a real diode (a few ohms instead of typically $10\,\Omega$ … $40\,\Omega$).

Shunt regulator



- the resistor takes the voltage difference: $I_R = (V_{\text{in}} - V_Z)/R$
- the Zener takes what the load does not need: $I_Z = I_R - I_L$
- the Zener must keep a minimum current $I_{Z,\text{min}}$ (about $5\,\text{mA}$, or $10\,\%$ of $I_{Z,\text{max}}$)

Designing a shunt regulator

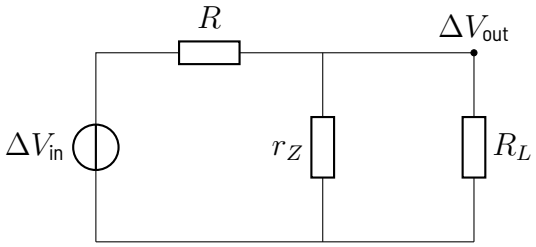
Design card: Zener shunt regulator

1. worst case for regulation: lowest V_{in}, largest load current $\Rightarrow\ R \le \dfrac{V_{\text{in,min}} - V_Z}{I_{L,\text{max}} + I_{Z,\text{min}}}$, choose the next **smaller** E-value
2. worst case for the Zener: highest V_{in}, no load $\Rightarrow\ I_{Z,\text{max}} = \dfrac{V_{\text{in,max}} - V_Z}{R}$, $P_Z = V_Z\, I_{Z,\text{max}}$
3. power of the resistor: $P_R = (V_{\text{in,max}} - V_Z)^2 / R$

- example: $12\,\text{V}$, $5.6\,\text{V}$, $I_L = 0\,\text{mA} \ldots 20\,\text{mA}$: $R \le 6.4\,\text{V}/25\,\text{mA} = 256\,\Omega \Rightarrow 220\,\Omega$
- the efficiency is poor: the full current flows all the time, even without load

▶ Falstad: `examples/sources/zener_shunt.txt`

Regulation and ripple



- for small changes the Zener is a resistor r_Z: small-signal equivalent circuit
- line regulation: $\dfrac{\Delta V_\text{out}}{\Delta V_\text{in}} = \dfrac{r_Z \parallel R_L}{R + r_Z \parallel R_L} \approx \dfrac{r_Z}{R}$
- load regulation: output resistance $R_\text{out} = R \parallel r_Z \approx r_Z$
- example $r_Z = 20\,\Omega$, $R = 220\,\Omega$: $1\,\text{V}$ ripple at the input $\rightarrow 83\,\text{mV}$ at the output

The Zener regulator suppresses input changes only by the ratio r_Z/R, typically a factor of 10. A larger resistor improves this, but requires a higher input voltage. Better regulation is obtained when the Zener diode is fed from a current source, or when a feedback loop with an amplifier is used (series regulator, voltage reference). The output resistance r_Z means that the output voltage changes with the load current: for $20\,\text{mA}$ and $r_Z = 20\,\Omega$ by $0.4\,\text{V}$.

Exercise 3.3: Design of a Zener shunt regulator

A load needs $5.6\,\text{V}$ and draws $0\,\text{mA}$ … $20\,\text{mA}$. The input voltage is $12\,\text{V}$. Use a $5.6\,\text{V}$ Zener diode with $I_{Z,\text{min}} = 5\,\text{mA}$.

a) Calculate the series resistor and choose an E12 value.

b) Calculate the maximum Zener current and the power of Zener diode and resistor. Which ratings do you choose?

c) Does the circuit still regulate at full load when the input drops to $11\,\text{V}$?

d) Read the output voltage without load and with $R_L = 270\,\Omega$ in Falstad and calculate I_Z.

Solution 3.3

a) $R \le (12\,\text{V} - 5.6\,\text{V})/(20\,\text{mA} + 5\,\text{mA}) = 256\,\Omega \Rightarrow 220\,\Omega$.

b) Without load: $I_{Z,\text{max}} = 6.4\,\text{V}/220\,\Omega = 29.1\,\text{mA}$, $P_Z = 5.6\,\text{V} \cdot 29.1\,\text{mA} = 163\,\text{mW} \Rightarrow 0.5\,\text{W}$ Zener (BZX79C5V6). $P_R = (6.4\,\text{V})^2/220\,\Omega = 186\,\text{mW} \Rightarrow 0.5\,\text{W}$ resistor.

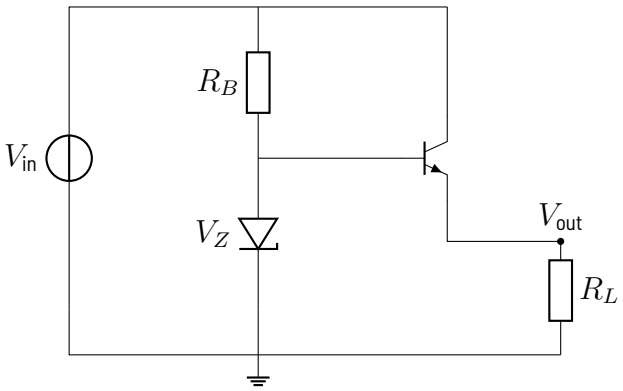
c) $I_R = (11\,\text{V} - 5.6\,\text{V})/220\,\Omega = 24.5\,\text{mA}$, $I_Z = 4.5\,\text{mA}$: just below $I_{Z,\text{min}}$; the voltage drops slightly, but the circuit still works. For a safe margin choose $200\,\Omega$ (E24).

d) ▶ Falstad: `examples/sources/zener_shunt.txt`
`Vout_open` $= 5.65\,\text{V}$ ($I_Z = I_R = 28.9\,\text{mA}$), `Vout_load` $= 5.61\,\text{V}$,
$I_L = 5.61\,\text{V}/270\,\Omega = 20.8\,\text{mA}$, $I_R = (12 - 5.61)\,\text{V}/220\,\Omega = 29.0\,\text{mA}$, $I_Z = 8.3\,\text{mA}$.

3.4 Series Regulators

The shunt regulator wastes current and cannot deliver large load currents. A series regulator puts a transistor between input and output. The transistor carries only the load current, and the Zener diode only has to deliver the small base current. Integrated linear regulators add an error amplifier, a bandgap reference and protection circuits; they are the standard solution for supplies up to about 1 A.

Emitter follower with Zener diode



Emitter follower regulator

$$V_{\text{out}} = V_Z - V_{BE} \approx V_Z - 0.65\,\text{V} \qquad I_B = \frac{I_L}{\beta + 1}$$

- the Zener sees only I_B: load changes hardly affect V_Z

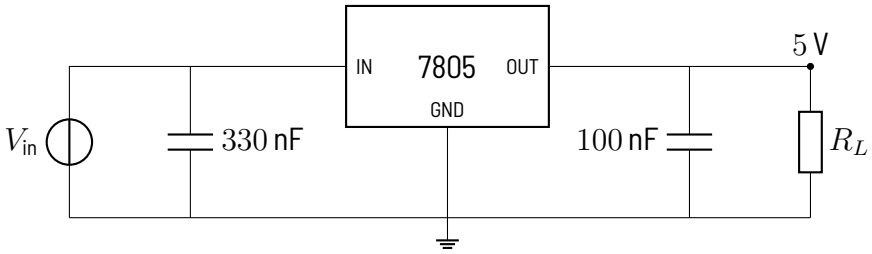
▶ Falstad: `examples/sources/series_regulator.txt`

Properties of the emitter follower regulator

- output resistance $\approx r_e + (R_B \parallel r_Z)/(\beta + 1)$ with $r_e = V_T/I_E$ ($0.25\,\Omega$ at $100\,\text{mA}$): much smaller than with the Zener alone
- V_{BE} depends on current and temperature ($-2\,\text{mV/K}$): V_{out} is only accurate to about $0.1\,\text{V}$
- transistor power: $P_Q = (V_{\text{in}} - V_{\text{out}}) \cdot I_L$ (heat sink!)
- design of R_B: $\dfrac{V_{\text{in,min}} - V_Z}{R_B} \geq I_{B,\text{max}} + I_{Z,\text{min}}$
- no short-circuit protection: a short at the output destroys the transistor

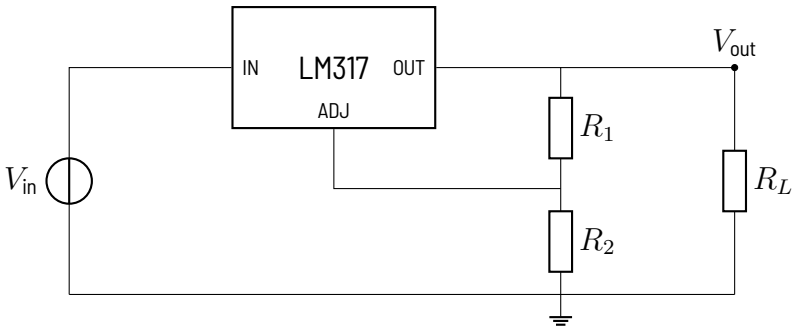
The Falstad example shows the effect: with $470\,\Omega$ (about $10\,\text{mA}$) the output is $4.95\,\text{V}$, with $47\,\Omega$ ($104\,\text{mA}$) it is $4.89\,\text{V}$. The Zener voltage itself changes by only $4\,\text{mV}$; most of the change comes from V_{BE}, which grows by $V_T \ln 10 = 60\,\text{mV}$ for ten times the current. An op-amp that compares the output with the reference and drives the transistor removes this error: that is the principle of the integrated regulator (section 3.5).

Integrated fixed regulator 78xx



- 7805, 7812, ...: fixed output voltage, $1\,\text{A}$ (TO-220), $0.1\,\text{A}$ (78L05); 79xx for negative voltages
- dropout about $2\,\text{V}$: $V_{\text{in}} \geq V_{\text{out}} + 2\,\text{V}$; LDO regulators: $0.1\,\text{V} \ldots 0.5\,\text{V}$
- internal current limit and thermal shutdown; capacitors close to the pins for stability

Adjustable regulator LM317



LM317

$$V_{\text{out}} = 1.25\,\text{V}\left(1 + \frac{R_2}{R_1}\right) + I_{\text{ADJ}}\,R_2 \qquad I_{\text{ADJ}} \approx 50\,\mu\text{A}$$

- the regulator keeps $1.25\,\text{V}$ between OUT and ADJ; $R_1 = 240\,\Omega$ sets $5.2\,\text{mA}$, enough for the minimum load

The LM317 has no ground pin. Its internal reference sits between OUT and ADJ, and all its quiescent current flows out of the output. The current through R_1 ($1.25\,\text{V}/R_1$) also flows through R_2; together with the small I_{ADJ} this gives the formula. The data sheet requires a minimum load current of about $10\,\text{mA}$ (typically $3.5\,\text{mA}$); the divider takes care of it. Without R_2 and with the load in place of R_2, the LM317 becomes a current source: $I = 1.25\,\text{V}/R_1$. Falstad has no model for the LM317; its principle (reference, error amplifier, series transistor) is the topic of the chapter on voltage references.

Exercise 3.4: Emitter follower regulator

A Zener diode $5.6\,\text{V}$ with $R_B = 1\,\text{k}\Omega$ drives the base of an npn transistor ($\beta = 100$) from $V_{\text{in}} = 12\,\text{V}$. The load is $R_L = 47\,\Omega$.

a) Estimate V_{out} and the load current with $V_{BE} = 0.7\,\text{V}$ (high current).

b) Calculate base current and Zener current.

c) Calculate the power of the transistor. Is a TO-92 transistor ($0.6\,\text{W}$) sufficient?

d) Which series resistor would a Zener shunt regulator need for the same load? Compare the idle current.

e) Compare with Falstad for $R_L = 47\,\Omega$ and $470\,\Omega$.

Solution 3.4

a) $V_{\text{out}} \approx 5.6\,\text{V} - 0.7\,\text{V} = 4.9\,\text{V}$, $I_L = 4.9\,\text{V}/47\,\Omega = 104\,\text{mA}$.

b) $I_B = 104\,\text{mA}/101 = 1.03\,\text{mA}$; $I_{R_B} = 6.4\,\text{V}/1\,\text{k}\Omega = 6.4\,\text{mA}$, $I_Z = 5.4\,\text{mA}$.

c) $P_Q = (12\,\text{V} - 4.9\,\text{V}) \cdot 104\,\text{mA} = 0.74\,\text{W}$: too much for TO-92; use a BD139 (TO-126) with a small heat sink.

d) $R \le 6.4\,\text{V}/(104\,\text{mA} + 5\,\text{mA}) = 59\,\Omega \Rightarrow 56\,\Omega$. Without load the Zener takes $114\,\text{mA}$ ($0.64\,\text{W}$); the series regulator only draws $6.4\,\text{mA}$.

e) ▶ Falstad: `examples/sources/series_regulator.txt`: $R_L = 47\,\Omega$: $V_{\text{out}} = 4.89\,\text{V}$, $I_L = 104\,\text{mA}$, $V_Z = 5.60\,\text{V}$; $R_L = 470\,\Omega$: $V_{\text{out}} = 4.95\,\text{V}$, $V_Z = 5.61\,\text{V}$.

Exercise 3.5: Adjustable regulator LM317

a) Design an LM317 regulator for $5\,\text{V}$ with $R_1 = 240\,\Omega$. Choose R_2 from E24 and calculate the actual output voltage (with $I_{\text{ADJ}} = 50\,\mu\text{A}$).

b) Which input voltage is needed at least (dropout $2\,\text{V}$)? What is the power of the LM317 at $12\,\text{V}$ input and $300\,\text{mA}$?

c) The LM317 is used as a current source for an LED string with $20\,\text{mA}$. Calculate the resistor.

Solution 3.5

a) $R_2 = R_1\,(5/1.25 - 1) = 3\,R_1 = 720\,\Omega \Rightarrow$ E24: $750\,\Omega$. $V_{\text{out}} = 1.25\,\text{V}\,(1 + 750/240) + 50\,\mu\text{A} \cdot 750\,\Omega = 5.16\,\text{V} + 0.04\,\text{V} = 5.20\,\text{V}$. (With E96: $R_2 = 715\,\Omega$ gives $4.97\,\text{V}$.)

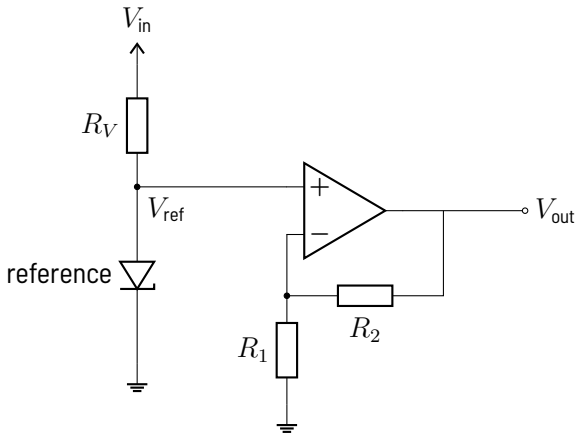
b) $V_{\text{in}} \geq 7.2\,\text{V}$. $P = (12\,\text{V} - 5.2\,\text{V}) \cdot 0.3\,\text{A} = 2.0\,\text{W}$: TO-220 with heat sink.

c) $R = 1.25\,\text{V}/20\,\text{mA} = 62.5\,\Omega \Rightarrow$ E24: $62\,\Omega$ ($20.2\,\text{mA}$); the LM317 then needs $V_{\text{in}} \geq V_{\text{LEDs}} + 1.25\,\text{V} + 2\,\text{V}$.

3.5 Op-Amp Voltage Sources

The emitter follower regulator loses about $0.1\,\text{V}$ between light and full load, because V_{BE} changes with the current. An op-amp removes this error. It compares the output voltage, divided by R_1 and R_2, with a reference voltage and drives the output until both are equal (golden rule 1, chapter 2). The reference only sees the op-amp input and is not loaded at all. The output voltage is set by a resistor ratio, and the output resistance is a few milliohms. For load currents above some $10\,\text{mA}$ a transistor inside the feedback loop boosts the output current. Reference, error amplifier and series transistor are exactly the blocks of every integrated linear regulator.

Reference and non-inverting amplifier



Op-amp voltage source

$$V_{\text{out}} = V_{\text{ref}}\left(1 + \frac{R_2}{R_1}\right) \qquad R_{\text{out}} \approx \frac{r_o}{1 + A\beta},\ \beta = \frac{R_1}{R_1 + R_2}$$

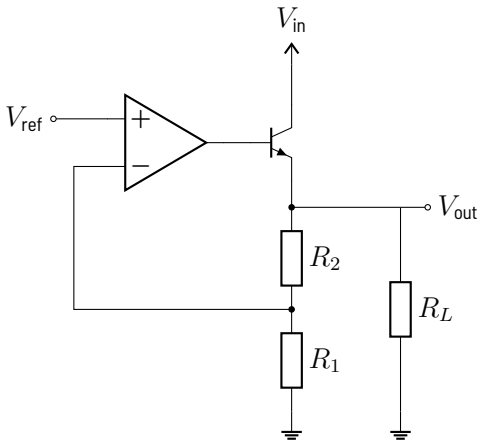
- the reference is not loaded; V_{out} is as accurate as V_{ref} and the ratio R_2/R_1

Output resistance and output current

- example: $r_o = 50\,\Omega$, $A = 10^5$, $R_1 = R_2$: $R_{\text{out}} = 50\,\Omega/(1 + 0.5 \cdot 10^5) = 1\,\text{m}\Omega$
  - 10 mA more load current changes V_{out} by only 10 µV
- but: the op-amp itself delivers only some 10 mA
- and: V_{out} must stay inside the output swing of the op-amp

The feedback makes the output resistance smaller by the loop gain $A\beta$, with the feedback factor β. This holds at low frequencies; at higher frequencies A falls (chapter 6) and the output resistance rises. A capacitor of some 100 nF at the load takes over at high frequencies. A large capacitor directly at the op-amp output can make it oscillate; data sheets state the allowed capacitive load.

More current: transistor inside the loop



- npn emitter follower as **booster**: the op-amp delivers only $I_B = I_L/(\beta + 1)$
- V_{BE} lies inside the loop: the op-amp output moves to $V_{\text{out}} + V_{BE}$, V_{out} stays exact

- transistor power $(V_{\text{in}} - V_{\text{out}}) \cdot I_L$; the op-amp output must reach $V_{\text{out}} + V_{BE}$

Example: $5.00\,\text{V}$ from $12\,\text{V}$, up to $100\,\text{mA}$

- reference LM4040-2.5 ($2.500\,\text{V}$, $0.1\,\%$) with $R_V = 4.7\,\text{k}\Omega$: $2.0\,\text{mA}$
- $G = 2$: $R_1 = R_2 = 10\,\text{k}\Omega$ ($0.1\,\%$) $\Rightarrow V_{\text{out}} = 5.00\,\text{V}$
- booster BD139 ($\beta \approx 100$): op-amp output $5.7\,\text{V}$ and $1\,\text{mA}$ at $100\,\text{mA}$ load

	$R_L = 470\,\Omega$	$R_L = 47\,\Omega$
emitter follower with Zener (section 3.4)	$4.95\,\text{V}$	$4.89\,\text{V}$
op-amp and booster	$5.000\,\text{V}$	$5.000\,\text{V}$

▶ Falstad: `examples/sources/opamp_voltage_source.txt`

In the Falstad example the reference is an ideal $2.5\,\text{V}$ source, which stands for the LM4040 with its series resistor. Both cells show `Vout_10mA` = `Vout_100mA` = $5.000\,\text{V}$, over the whole range of the R_L slider ($20\,\Omega$ to $1\,\text{k}\Omega$). The transistor dissipates $(12\,\text{V} - 5\,\text{V}) \cdot 106\,\text{mA} = 0.74\,\text{W}$ at $47\,\Omega$, the same as in the emitter follower regulator. An integrated regulator adds what this circuit lacks: a current limit and a thermal shutdown.

Exercise 3.6: Op-amp voltage source for $3.3\,\text{V}$

A $3.3\,\text{V}$ supply for a sensor board is built from a $2.5\,\text{V}$ reference, an op-amp and an npn booster ($\beta = 100$, $V_{BE} = 0.75\,\text{V}$). The input voltage is $12\,\text{V}$, the load current up to $200\,\text{mA}$.

a) Choose R_2 for $R_1 = 10\,\text{k}\Omega$ from E24 and from E96. Calculate the output voltages.

b) Which voltage and current must the op-amp output deliver at full load?

c) Calculate the power of the transistor. Compare with the power delivered to the load.

d) In the Falstad example, turn the R_L slider from $1\,\text{k}\Omega$ to $20\,\Omega$. What happens to the output voltage, and what would limit the output current in a real circuit?

Solution 3.6

a) $R_2 = R_1\,(3.3/2.5 - 1) = 3.2\,\text{k}\Omega$. E24: $3.3\,\text{k}\Omega$, $V_{\text{out}} = 2.5\,\text{V} \cdot 1.33 = 3.325\,\text{V}$ ($0.8\,\%$); E96: $3.24\,\text{k}\Omega$, $V_{\text{out}} = 3.310\,\text{V}$ ($0.3\,\%$).

b) $V_{\text{out}} + V_{BE} \approx 4.06\,\text{V}$; $I_B = 200\,\text{mA}/101 = 1.98\,\text{mA}$ (the divider takes only $0.25\,\text{mA}$ more from the emitter).

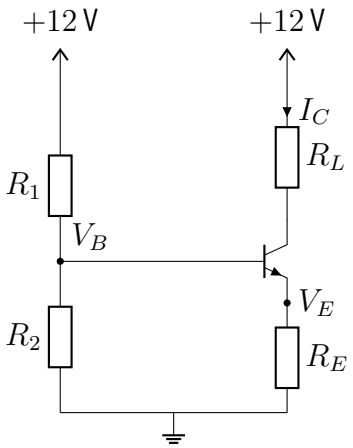
c) $P_Q = (12\,\text{V} - 3.31\,\text{V}) \cdot 0.2\,\text{A} = 1.74\,\text{W}$, the load gets only $0.66\,\text{W}$: efficiency $28\,\%$. A heat sink is needed; a switching regulator would be the better choice for this current.

d) ▶ Falstad: `examples/sources/opamp_voltage_source.txt`: `Vout_100mA` stays at $5.000\,\text{V}$ over the whole slider range ($250\,\text{mA}$ at $20\,\Omega$). The simulation has no current limit; in reality the transistor power ($1.75\,\text{W}$), its safe operating area and the op-amp output current (I_L/β) limit the current.

3.6 Transistor Current Sources

A bipolar transistor in its active region is a current source by nature: its collector current depends on the base–emitter voltage, but hardly on the collector voltage. An emitter resistor turns this into a current source with a well-defined current. Inside integrated circuits the current mirror is the most important building block; there is hardly an op-amp or a comparator without several of them.

BJT current source



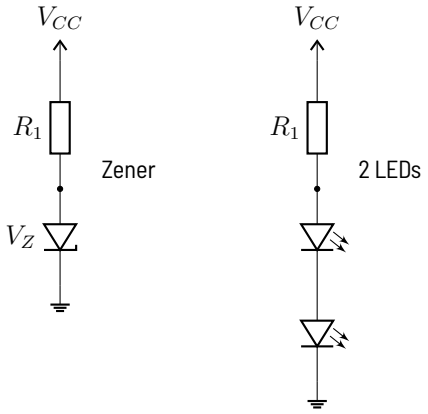
BJT current source

$$I_C \approx I_E = \frac{V_B - V_{BE}}{R_E} \approx \frac{V_B - 0.65\,\text{V}}{R_E}$$

- the emitter follows the base: $V_E = V_B - V_{BE}$ is fixed, so the emitter current is fixed
- the load sits in the collector and may vary within the compliance range

Compliance range and base voltage

- the transistor must stay active: $V_{CE} \geq$ about $0.3\,\text{V}$, i.e. $V_C \geq V_E + 0.3\,\text{V}$
- compliance: $R_{L,\text{max}} = \dfrac{V_{CC} - V_E - 0.3\,\text{V}}{I_C}$; a larger load saturates the transistor and the current drops
- stiff base voltage: divider current $\geq 10\,I_B$; base current error $I_B \cdot (R_1 \parallel R_2)$
- temperature: V_{BE} drops by $2\,\text{mV/K}$; a large V_E (1 V and more) makes the current stable



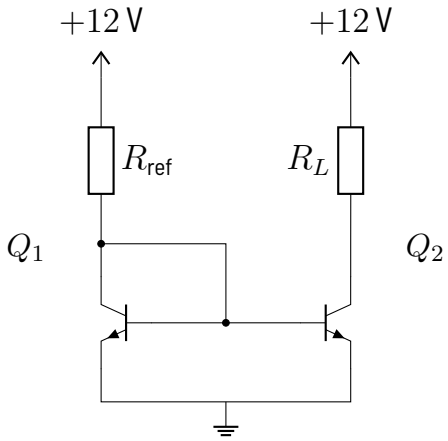
The base voltage can come from a divider, from a Zener diode or from forward-biased diodes or LEDs. The diode solutions make V_B independent of the supply voltage. Two silicon diodes ($V_B \approx 1.3\,\text{V}$) have the additional advantage that their temperature coefficient cancels that of V_{BE}: the voltage across R_E is one diode drop and stays constant. A red LED gives $V_B \approx 1.8\,\text{V}$ and $V_E \approx 1.15\,\text{V}$, again with partial temperature compensation.

Example: $1\,\text{mA}$ current source

- $V_{CC} = 12\,\text{V}$, $R_1 = 10\,\text{k}\Omega$, $R_2 = 2.2\,\text{k}\Omega$: $V_B = 2.16\,\text{V}$ (unloaded)
- $R_E = 1.5\,\text{k}\Omega$: $I_E = (2.16\,\text{V} - 0.65\,\text{V})/1.5\,\text{k}\Omega = 1.0\,\text{mA}$
- loads $1\,\text{k}\Omega$ and $4.7\,\text{k}\Omega$: the same current; $15\,\text{k}\Omega$ is outside the compliance range

▶ Falstad: `examples/sources/bjt_current_source.txt` The simulation gives $1.023\,\text{mA}$ for both the $1\,\text{k}\Omega$ and the $4.7\,\text{k}\Omega$ load. The difference to the hand calculation comes from V_{BE}: the Falstad transistor model has $V_{BE} = 0.60\,\text{V}$ at $1\,\text{mA}$, real small signal transistors (BC547) are closer to $0.65\,\text{V}$. With $15\,\text{k}\Omega$ the collector would have to go down to $12\,\text{V} - 15\,\text{V} < 0$; the transistor saturates and the current falls to $0.71\,\text{mA}$. The Falstad model has no Early effect, so in the active region the current does not depend on the load at all. A real transistor has an output resistance of some $100\,\text{k}\Omega$, increased by the emitter resistor to several $\text{M}\Omega$.

Current mirror



- Q_1 is connected as a diode: it sets V_{BE} for the reference current
- Q_2 has the same V_{BE} and therefore the same collector current: $I_{\text{out}} = I_{\text{ref}}\,\beta/(\beta + 2)$
- $I_{\text{ref}} = (V_{CC} - V_{BE})/R_{\text{ref}}$; with $11\,\text{k}\Omega$: $1.04\,\text{mA} \rightarrow 1.02\,\text{mA}$

▶ Falstad: `examples/sources/current_mirror.txt` The current mirror works only if both transistors are identical and at the same temperature, so it is used in integrated circuits or with matched transistor pairs (BCM847). With discrete transistors emitter resistors of a few $100\,\Omega$ improve the matching. The output current changes with the collector voltage because of the Early effect; the Wilson mirror and cascode mirrors reduce this. Mirrors with several output transistors distribute one reference current to many stages; by scaling the emitter areas, the output current becomes a multiple of the reference current.

Exercise 3.7: BJT current source for a resistive sensor

A resistive sensor ($R_S = 1.0\,\text{k}\Omega \ldots 1.4\,\text{k}\Omega$) shall be excited with $1\,\text{mA}$ by a BJT current source at $V_{CC} = 12\,\text{V}$ ($V_{BE} = 0.65\,\text{V}$, $\beta = 100$).

a) The base voltage comes from $R_1 = 10\,\text{k}\Omega$, $R_2 = 2.2\,\text{k}\Omega$. Calculate V_B with and without the base current, and choose R_E from E12.

b) Calculate the voltage across the sensor at $1.0\,\text{k}\Omega$ and $1.4\,\text{k}\Omega$, and the compliance range.

c) The base divider is replaced by two red LEDs in series ($V_F = 1.8\,\text{V}$ each) fed by $4.7\,\text{k}\Omega$. Recalculate R_E.

d) Read the collector currents for $1\,\text{k}\Omega$, $4.7\,\text{k}\Omega$ and $15\,\text{k}\Omega$ in Falstad.

Solution 3.7

a) Unloaded: $V_B = 12\,\text{V} \cdot 2.2/12.2 = 2.16\,\text{V}$. $I_B \approx 10\,\mu\text{A}$ through $R_1 \parallel R_2 = 1.8\,\text{k}\Omega$ lowers it by $18\,\text{mV}$: $V_B = 2.15\,\text{V}$. $R_E = (2.15\,\text{V} - 0.65\,\text{V})/1\,\text{mA} = 1.5\,\text{k}\Omega$ (E12).

b) $V_S = 1.0\,\text{V}$ and $1.4\,\text{V}$. With $V_E = 1.5\,\text{V}$ and $V_{CE} \geq 0.3\,\text{V}$: $R_{L,\text{max}} = (12 - 1.5 - 0.3)\,\text{V}/1\,\text{mA} = 10.2\,\text{k}\Omega$; the sensor is far inside.

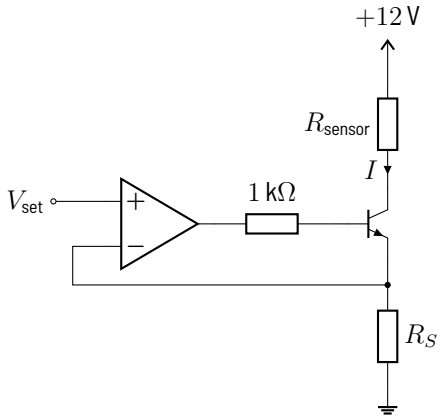
c) $V_B = 3.6\,\text{V}$, $R_E = (3.6\,\text{V} - 0.65\,\text{V})/1\,\text{mA} = 2.95\,\text{k}\Omega \Rightarrow$ E24 $3.0\,\text{k}\Omega$ ($0.98\,\text{mA}$). V_B no longer depends on V_{CC}.

d) ▶ Falstad: `examples/sources/bjt_current_source.txt`: $V_B = 2.15\,\text{V}$, $V_E = 1.55\,\text{V}$, $I_C = 1.023\,\text{mA}$ for $1\,\text{k}\Omega$ and $4.7\,\text{k}\Omega$ ($V_C = 7.19\,\text{V}$), $0.71\,\text{mA}$ for $15\,\text{k}\Omega$ (saturated).

3.7 Op-Amp Current Sources and Sensor Excitation

The BJT current source depends on V_{BE} and β. An op-amp in the loop removes the V_{BE} error completely: it forces the voltage across the sense resistor to equal a set voltage. Such voltage-to-current converters are used for sensor excitation, for $4\,\text{mA} \ldots 20\,\text{mA}$ current loops and as programmable current sinks. As for the voltage source, we only need the golden rules (chapter 2): an op-amp with negative feedback makes the voltage between its inputs zero.

Op-amp and transistor

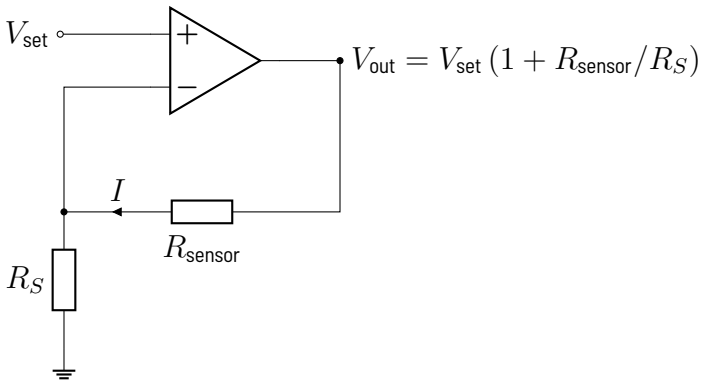


Op-amp current source

$$V_{R_S} = V_{\text{set}} \quad \Rightarrow \quad I_E = \frac{V_{\text{set}}}{R_S} \qquad I = I_C = I_E \, \frac{\beta}{\beta + 1}$$

- V_{BE} and its drift are inside the loop and drop out; the base current is lost ($1\,\%$ at $\beta = 100$, a MOSFET avoids it)

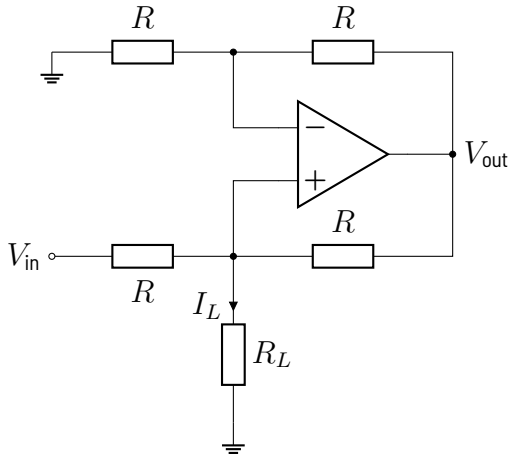
Floating load in the feedback path



- non-inverting amplifier: the current V_{set}/R_S through R_S must flow through the sensor
- exact current (no base current); but neither sensor terminal is at ground
- compliance: V_{out} must stay inside the op-amp output range

▶ Falstad: `examples/sources/opamp_current_source.txt`

Howland current pump (grounded load)



- four equal resistors: $I_L = V_{in}/R$, independent of R_L (within the compliance range)
- the resistors must be matched well: a mismatch of $0.1\,\%$ limits the output resistance

With four equal resistors the inverting input is at $V_{out}/2$. Because the op-amp makes both inputs equal, the load voltage is $V_L = V_{out}/2$. KCL at the non-inverting input gives $I_L = (V_{in} - V_L)/R + (V_{out} - V_L)/R = V_{in}/R$: the load voltage cancels. The Howland circuit is elegant but sensitive to resistor tolerances; integrated difference amplifiers with laser-trimmed resistors (e.g. INA105) are a good basis for it.

Sensor excitation: why 1 mA?

- resistive temperature sensor (Pt1000 like): $R(T) = R_0\,(1 + \alpha T)$, $R_0 = 1\,\text{k}\Omega$, $\alpha = 3.85 \times 10^{-3}/\text{K}$
  - $0\,°\text{C}$: $1000\,\Omega$, $100\,°\text{C}$: $1385\,\Omega$
- with a constant current I: $V_{sensor} = I \cdot R(T)$ is linear in T: $3.85\,\text{mV/K}$ at $1\,\text{mA}$
- more current gives more signal, but **self-heating**: $\Delta T = P/G_{th}$
  - $P = I^2R = (1\,\text{mA})^2 \cdot 1.385\,\text{k}\Omega = 1.39\,\text{mW}$; with a typical $0.4\,\text{K/mW}$: $0.55\,\text{K}$ error
- the signal offset ($1\,\text{V}$ at $0\,°\text{C}$) is removed by a **bridge**

The power in the sensor heats it above the temperature it should measure. The self-heating coefficient depends strongly on the mounting: a thin-film sensor in still air heats up by some tenths of a kelvin per milliwatt, in flowing water ten times less. Since the power grows with the square of the current, halving the current reduces the error to one quarter, but also halves the signal. $1\,\text{mA}$ for a $1\,\text{k}\Omega$ sensor (and $0.3\,\text{mA}$ for a Pt1000 in precise applications, $1\,\text{mA}$ for a Pt100) are common compromises.
A current source makes the sensor voltage proportional to its resistance, but the large constant part (IR_0) remains. The capstone task therefore uses a Wheatstone bridge: a second branch with fixed resistors produces the voltage that the sensor has at the lower end of the range, and the instrumentation amplifier only amplifies the difference. The bridge is fed from a stable voltage, and its resistors are chosen so that the sensor current stays below $1\,\text{mA}$. The chapter on bridge circuits treats this in detail.

Example: 1 mA for a sensor

- $V_{set} = 1.00\,\text{V}$ (from a reference), $R_S = 1.00\,\text{k}\Omega$ ($0.1\,\%$): $I = 1.000\,\text{mA}$
- op-amp and transistor: sensor current $0.990\,\text{mA}$ (base current missing)
- floating sensor in the feedback path: $1.000\,\text{mA}$, $V_{out} = 1\,\text{V}\,(1 + 1.385) = 2.385\,\text{V}$ at $100\,°\text{C}$
- accuracy of the current = accuracy of V_{set} and R_S

▶ Falstad: `examples/sources/opamp_current_source.txt` Falstad shows exactly these numbers: VS $= 1.000$ V, the sensor in the collector carries 0.990 mA, the floating sensor 1.000 mA with VOUT2 $= 2.385$ V. The $1\,\mathrm{k}\Omega$ resistor between op-amp and base limits the output current of the op-amp and decouples the capacitive load of the base. For precise sensor measurements the ratiometric principle is often even better than a precise current: the same reference voltage that sets the current is also used as the reference of the ADC, so its errors cancel.

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